

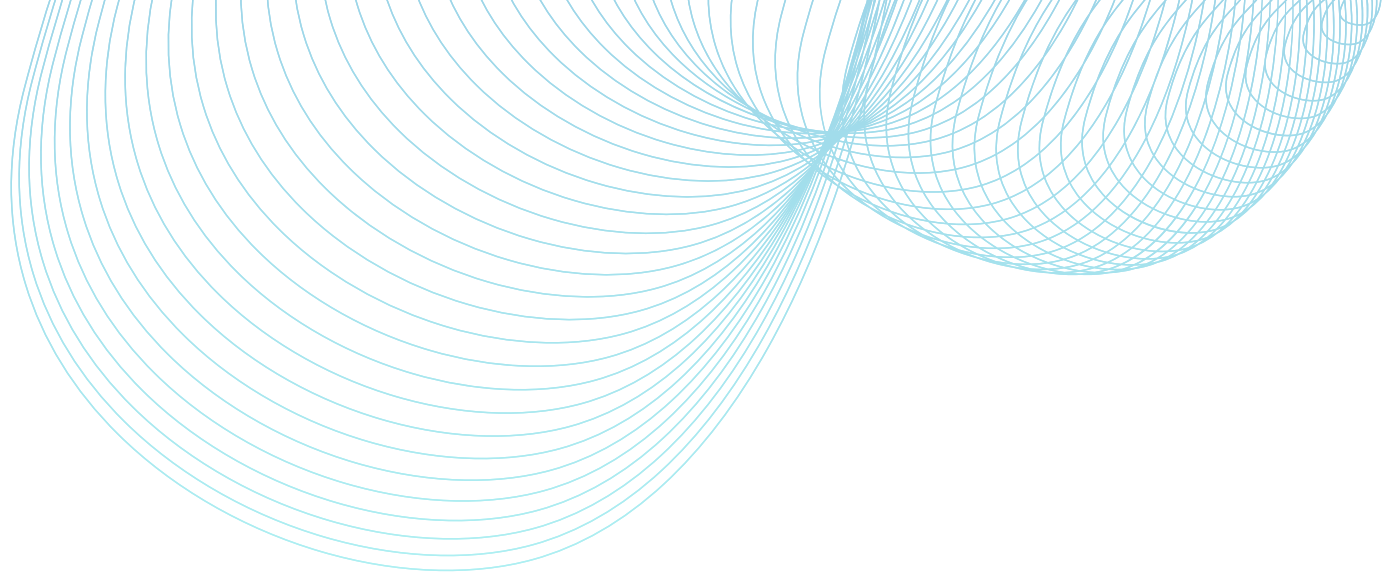


TEAM - 5
WEEK 3

Leveraging LLMs to Understand Global Mental Health Well-being & Fomo in Social Media

AGENDA

- Sentiment Analysis
- Pattern Identification
- Correlation Analysis
- Architecture
- Challenges
- Focus areas for next week





1

Correlation Analysis

Victoria Voloshyna

The purpose of this analysis was to identify correlations between variables that may potentially have an impact on mental health.

2

Pattern Identification

Elizabeth Waithaka

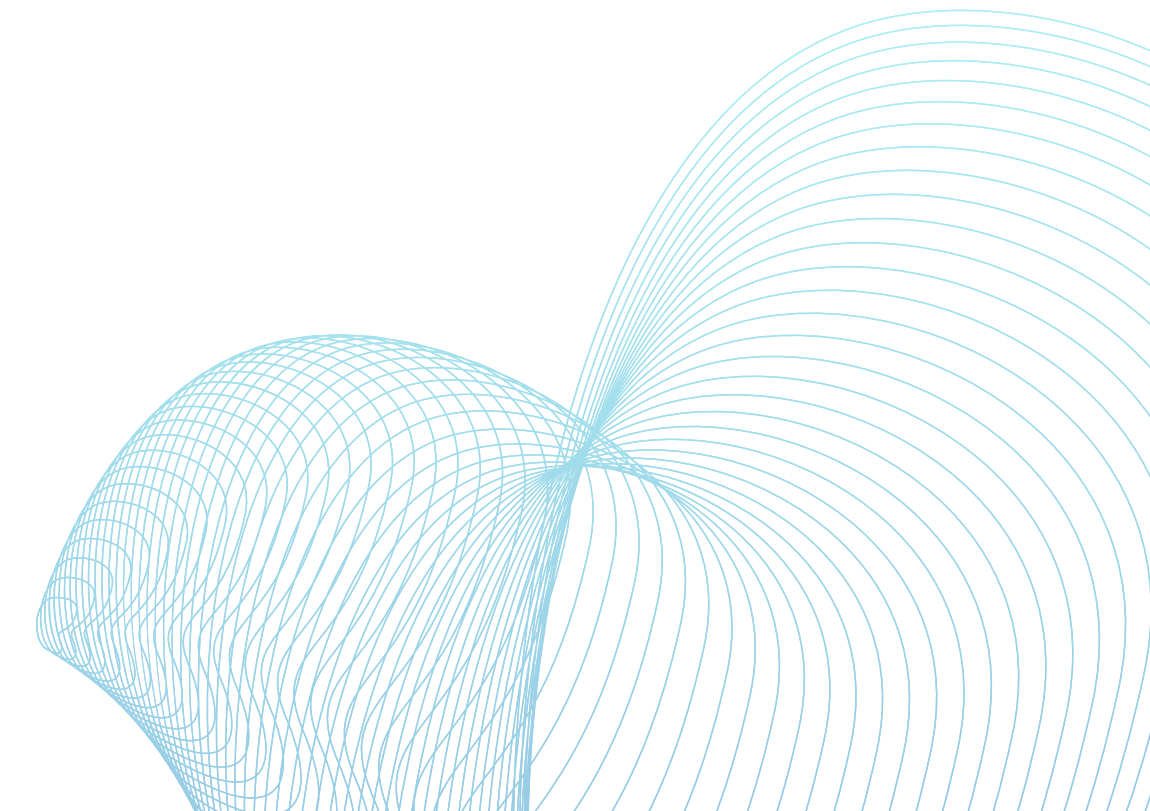
Analysing vast amounts of text data to find recurring themes or patterns.

3

Sentiment Analysis

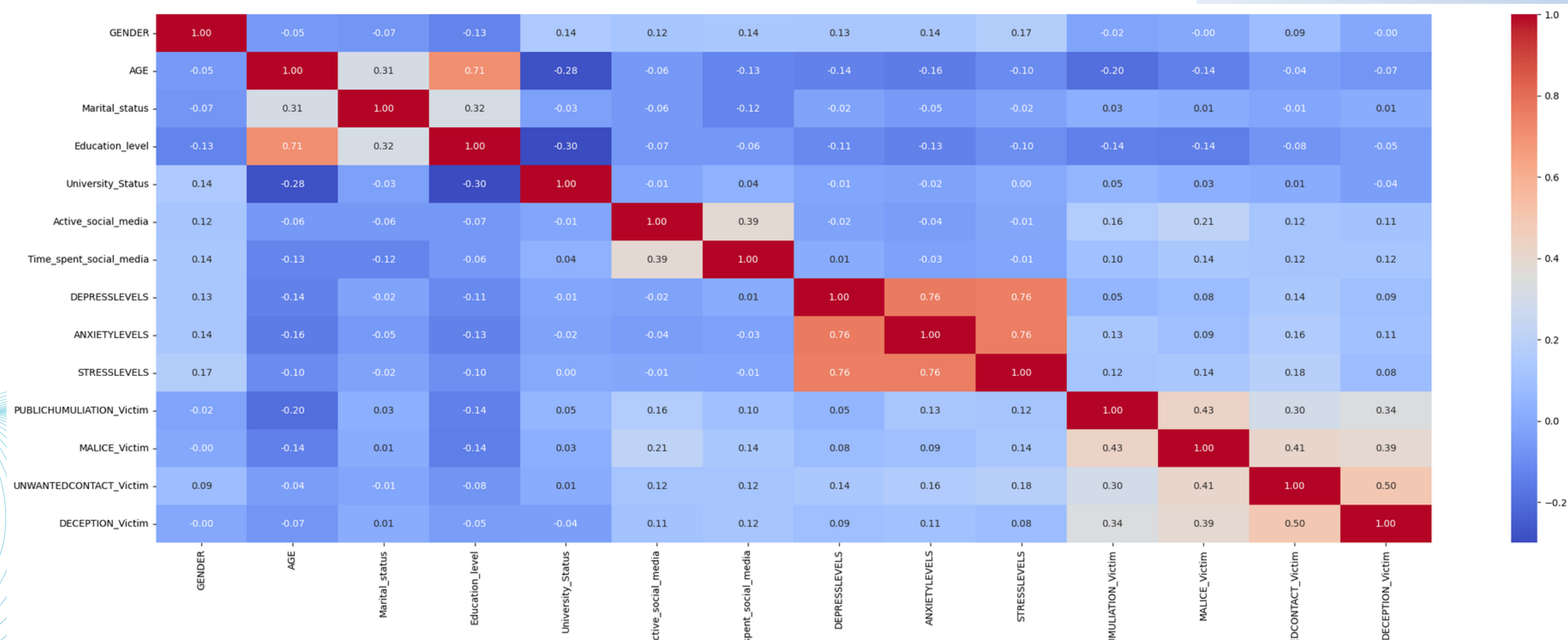
Pranav Gavali

Sentiment analysis involves examining large amounts of text data (like social media posts, etc) to understand the overall emotional tone.



CORRELATION ANALYSIS

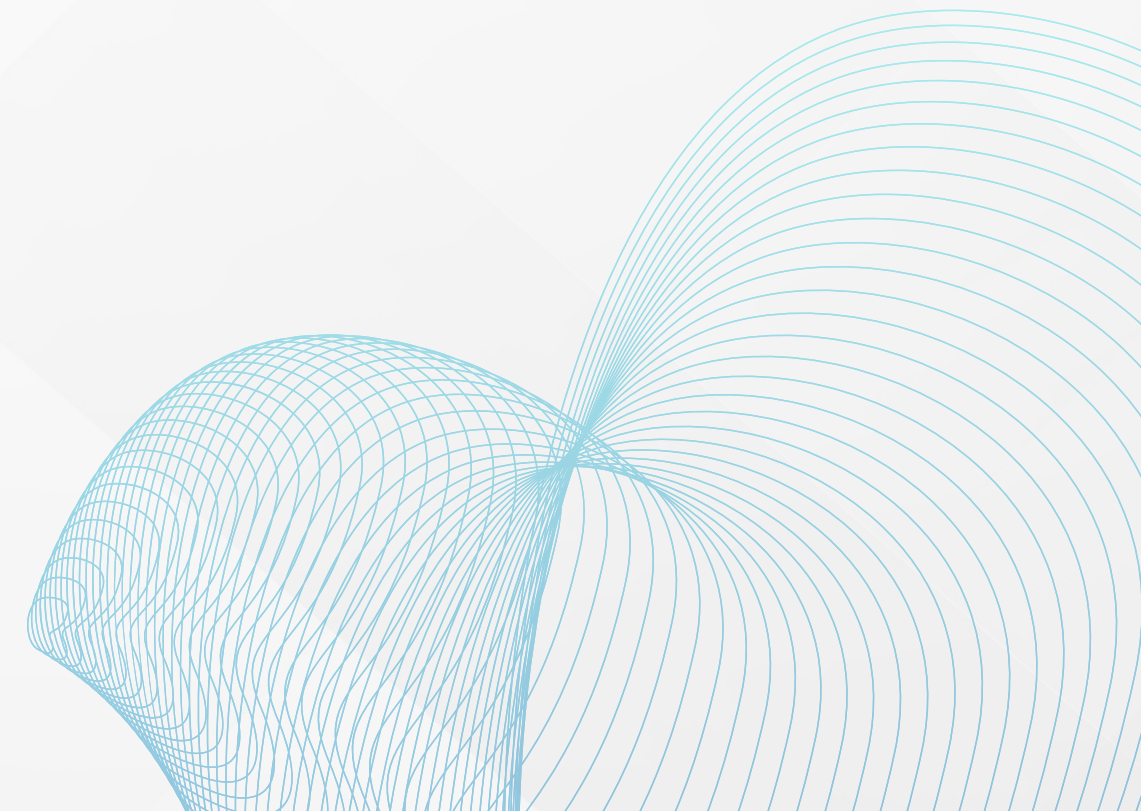
- There is high positive correlation between depression levels and Anxiety, Stress Levels.
- There is positive correlation of depression levels and Gender (more depressed females), Victims of unwanted contact on social media.
- There is negative correlation of depression levels and Age, Education level.



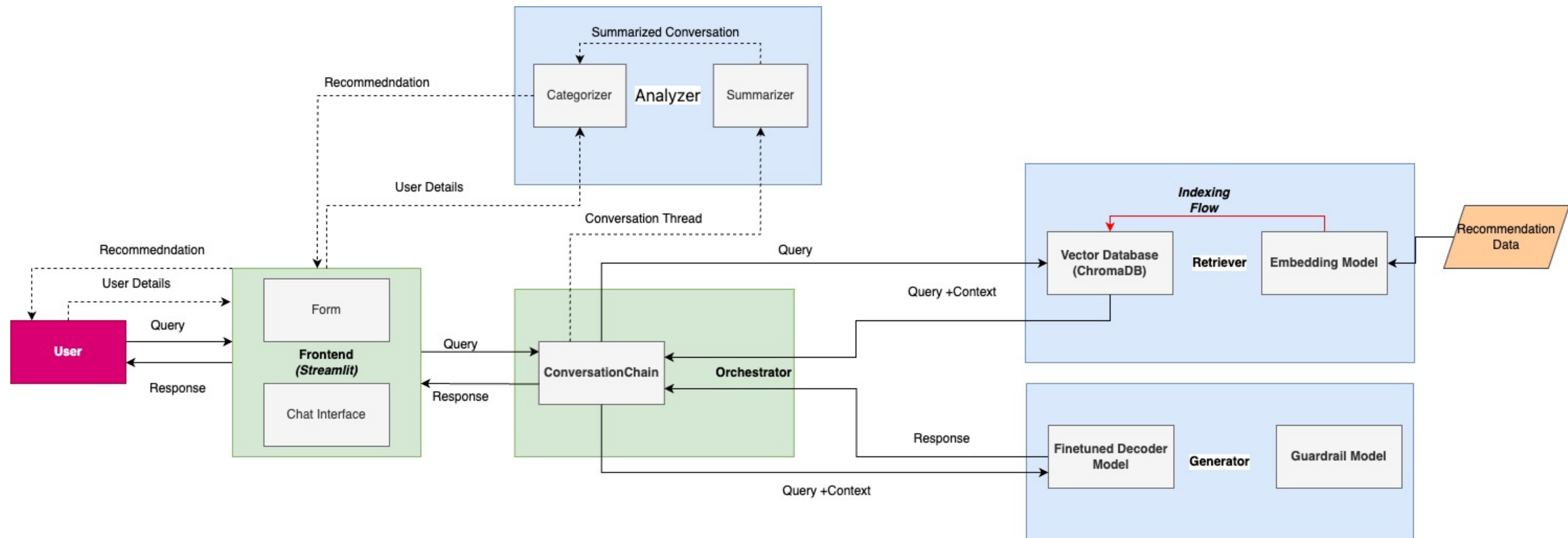
CODE WALKTHROUGH

PATTERN IDENTIFICATION - ELIZABETH WAITHAKA

SENTIMENT ANALYSIS - DEBANJAN MONDAL



ARCHITECTURE



OVERALL SOLUTION DEMO

- UI (Streamlit)
- RAG, VectorDB and LLM integration
- Huggingface deployment

FINE TUNING AND ALIGNMENT

Sayak Mallick

Supervised Fine-Tuning (SFT)

Aligning LLMs with Direct Preference Optimization (DPO)

CURRENT CHALLENGES :

- **GPU availability for improved response and testing**
- **RASA and StreamLit integration**
- **Frequently updated frameworks and Libraries made difficult to learn from existing code snips**
- **Vector Embeddings of CSV files**
- **Working with combination of LLamaIndex and LangChain**



FOCUS AREAS FOR NEXT WEEK

- **Ingesting Conversational data (CSV) into VectorDB**
- **Advanced RAG and User Experience**
- **Advance Fine-tuning / DPO**
- **LLM Guardrail Integration**
- **Testing and Validation**
- **Deployment Tuning**

Acknowledgements

PRODUCT MANAGER: KRISHNA KISHORE GAJJALA

CO- PM: ANKITHA BEERAM

MODEL BUILDING AND UI/UX TRACK

Lead

Debanjan Mondal

Phaneendra Ganji

Co Lead

Pranav Gavali

Collaborators

Meriem Baziz

Vijaya Krishna

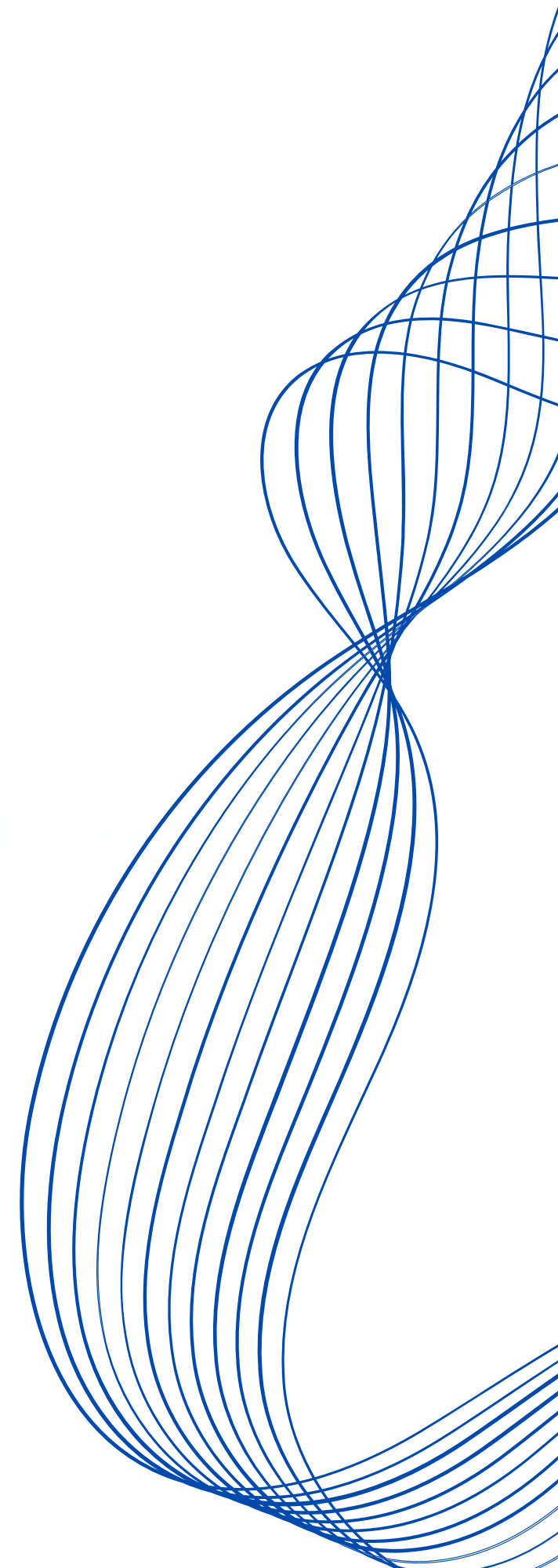
Victoria Voloshyna

Kỳ Nguyễn

Amos Fang

Sayak Mallick

Elizabeth Waithaka



**THANK
YOU!**

